

US EPA TOXCAST DATA RELEASE AUGUST 2025 DATABASE & tcpl R PACKAGE

This file describes the contents of the August 2025 ToxCast data release. The MySQL data consists of the following files:

`invitrodb_v4_3_mysql.gz`

Database Installation Overview

This file contains a dump of the invitrodb v4.3 MySQL database. The database is large, containing over half a billion records, and is provided in a compressed format (`invitrodb_v4_3_mysql.sql.gz`).

To utilize the database, the user must have access to a MySQL server and load the database contained in the `invitrodb_v4_3_mysql.gz` file. The file is a zipped SQL file containing the entire Invitrodb database. To avoid errors while calling large queries, we recommend the MySQL server setting, `max_allowed_packet`, is set to at least 1 GB.

- Download and Decompression

1. Download the `invitrodb_v4_3_mysql.sql.gz` file to your local machine.
2. Decompress the file using `gzip`:

```
gunzip invitrodb_v4_3_mysql.sql.gz
```

This will create an uncompressed file named `invitrodb_v4_3_mysql.sql`.

- Importing the Database

 To import/restore or load the database in MySQL:

1. Ensure you have a MySQL server running and have appropriate permissions to create and manipulate databases.
2. Create a new database for importing the data:

```
CREATE DATABASE invitrodb_v4_3;
```

3. Import the data into the new database. Given the size of the database, this process can be run in the background to avoid blocking other operations. This can be achieved by appending an ampersand (&) at the end of the command:

```
mysql --user=yourusername --password=yourpassword --host=yourhost invitrodb_v4_3  
< invitrodb_v4_3_mysql.sql &
```

Please note that the time taken to complete the restoration process can vary significantly depending on the resources and settings of your database server.

4. You can check the progress of the import using the following command:

```
show processlist;
```

Run this command in a new MySQL shell. The State column will give you an idea of what the server is currently working on.

Remember to replace `yourusername`, `yourpassword`, and `yourhost` with your actual MySQL username, password, and host information. Always ensure that your MySQL server has enough resources to handle the restoration of large databases. If necessary, consider adjusting your server settings to optimize the import process. More comprehensive documentation for installing MySQL and creating a database from SQL dump files can be found online at the MySQL Documentation: <https://dev.mysql.com/doc/>.

tcpl Package Installation

The R software package `tcpl` is available on CRAN <https://cran.r-project.org/web/packages/tcpl/index.html> and GitHub <https://github.com/USEPA/CompTox-ToxCast-tcpl>. Given `tcplLite` is no longer supported by `tcpl`, a MySQL connection is required for data processing. When `tcpl` is loaded, the default configuration sets the options to `tcpl`'s application API key to support new users testing out the package. This default API key is not intended for regular users; instead, it is highly recommended to obtain a personal API key to also access other CTX APIs. For this, send an email request to CTX API support at ccte_api@epa.gov. For standalone curve-fitting without connectivity to the MySQL database `invitrodb`, see the R software package dependency `tcplfit2`, on CRAN <https://cran.r-project.org/web/packages/tcplfit2/index.html> and GitHub <https://github.com/USEPA/CompTox-ToxCast-tcplFit2>.

To install `tcpl`, the user will first need to properly install Rtools, enabling R to compile the source package. The v3.3.0 package relies on R ($\geq 4.1.0$), and the following packages: `data.table` ($\geq 1.15.0$), `DBI`, `RMariaDB`, `numDeriv`, `RColorBrewer`, `utils`, `stats`, `methods`, `graphics`, `grDevices`, `sqldf`, `dplyr`, `tidyr`, `plotly`, `tcplfit2`, `ggplot2`, `gridExtra`, `stringr`, `rlang`, `ctxR` ($\geq 1.1.0$), `viridis`, `gt`. To interact with the MySQL database, the user may consider using the `RMariaDB` package in addition to `tcpl` functions.

The package can be installed from CRAN using the following R command:

```
install.packages("tcpl")
```

For the most current development version of `tcpl`, run the following R command:

```
devtools::install_github("USEPA/CompTox-ToxCast-tcpl")
```

Check out the vignette for the ToxCast Data Analysis Pipeline (`tcpl`) R package for more comprehensive documentation describing ToxCast data processing, retrieval, and interpretation. For additional questions or concerns, please contact Madison Feshuk (feshuk.madison@epa.gov).